02-BASIC-ELECTRONIC-COMPONENTS-AND-ICS

Item #	Discrete Component Name	Technical Specifications & Model Details	Practical Lab / Circuit Role	Required Quantity	Asset Classification
01	**Bare 5V Sugar-Cube Electromagnetic Relay**	Songle SRD-05VDC-SL-C; 5-pin SPDT bare relay; contacts rated 10A 250V AC / 30V DC *(Requires manual transistor & diode driver)*	Learning coil actuation, transistor switching & flyback diode protection	**5 Units**	**Consumable**
02	**Single 1-Digit 7-Segment LED Display (Common Cathode)**	0.56-inch 1-digit red 7-segment display; 10-pin DIP through-hole; forward voltage 2.0V *(Requires 7-8 current-limiting resistors)*	Learning direct binary-to-decimal LED segment decoding	**5 Units**	**Consumable**
03	**Single 1-Digit 7-Segment LED Display (Common Anode)**	0.56-inch 1-digit red 7-segment display; 10-pin DIP through-hole; active-low configuration	Comparative study of active-high vs active-low segment logic	**5 Units**	**Consumable**
04	**Bare 4-Digit 7-Segment LED Display**	12-pin bare multiplexed 4-digit display *(No TM1637 driver IC; requires multiplexed pin scanning)*	Teaching display multiplexing, ghosting prevention & persistence of vision	**5 Units**	**Consumable**
05	**Bare 16×2 Character LCD Panel (HD44780)**	16-pin parallel interface LCD display; 5V DC; blue backlight *(Raw display without I2C backpack board)*	Teaching parallel data buses (4-bit/8-bit), RS, Enable & RW pin control	**5 Units**	**Consumable**
06	**10-Segment LED Bargraph Display Array**	20-pin DIP array with 10 individually addressable red/green LED segments	Analog level indicators, battery level meters & audio VU meters	**5 Units**	**Consumable**
07	**Bare LDR Photoresistors (GL5528)**	5mm cadmium sulfide (CdS) light-dependent resistor; dark resistance 1 MΩ, light resistance 10-20 kΩ	Voltage divider circuits for automatic light/dark detectors	**5 Units**	**Consumable**
08	**Bare NTC Thermistors (10kΩ Bead)**	2-pin negative temperature coefficient thermistor; B(25/50) = 3950 K; resistance decreases as heat rises	Building analog thermal detection and fire alarm circuits	**5 Units**	**Consumable**
09	**Rotary Carbon Potentiometers (10kΩ)**	3-pin panel mount / breadboard rotary potentiometer with knurled shaft and matching control knobs	Dial controls for PWM motor speed, tone pitch & LCD contrast	**5 Units**	**Consumable**
10	**Breadboard Miniature Trimpots (10kΩ - 3362P)**	3-pin single-turn horizontal/vertical cermet trimmer potentiometer	Precision threshold calibration on prototype breadboards	**5 Units**	**Consumable**
11	**ATmega328P-PU Standalone IC (with 16MHz Crystal Kit)**	28-pin DIP AVR microcontroller bare chip paired with 16 MHz HC-49S quartz crystal and two 22pF ceramic load capacitors	Building a "DIY Arduino on Breadboard / Perfboard" from scratch	**5 Sets**	**Consumable**
12	**ATmega16U2 Standalone IC**	8-bit AVR microcontroller bare IC in DIP/SOIC package; 16 KB flash; USB controller capability; 2.7V - 5.5V	Advanced USB microcontroller prototyping & bare IC design	**3 Units**	**Consumable**
13	**Shift Register IC (74HC595)**	16-pin DIP 8-bit serial-in parallel-out shift register with 3-state output latches	Controlling 8 to 64 LEDs or multi-digit displays using only 3 GPIO pins	**5 Units**	**Consumable**
14	**Dual Operational Amplifier IC (LM358)**	8-pin DIP dual low-power general-purpose op-amp; single supply 3V - 32V	Building analog signal amplifiers, active filters & sensor preamps	**5 Units**	**Consumable**

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15	**Dual Voltage Comparator IC (LM393)**	8-pin DIP dual precision voltage comparator; open-collector output	Understanding analog voltage threshold slicing (how sensor modules work)	**5 Units**	**Consumable**
16	**Timer IC — LM555 / NE555**	Precision timing bipolar IC; DIP-8 package; operating voltage 4.5V - 16V	Astable multivibrators (clock pulse generators) and monostable timers	**5 Units**	**Consumable**
17	**Linear Voltage Regulators (LM78XX)**	3-terminal linear regulators in TO-220 package: LM7805 (+5V), LM7809 (+9V), LM7812 (+12V); 1.5A peak	Designing clean regulated DC power supply rails on breadboards	**5 Units**	**Consumable**
18	**Logic-Level Power N-Channel MOSFET (IRLZ44N / IRF540N)**	TO-220 power MOSFET; 55V, 47A, low R(DS(on)); logic-level gate threshold (V(GS(th)) = 1.0V - 2.0V)	Direct high-speed switching of high-current DC motors, coils & 12V LED strips	**5 Units**	**Consumable**
19	**Bare Dual H-Bridge Motor Driver IC (L293D)**	16-pin DIP bare driver IC (unmounted without shield); 600mA per channel; internal clamp diodes	Breadboard motor control circuits and H-bridge theory	**5 Units**	**Consumable**
20	**Bare Darlington Transistor Array IC (ULN2003A)**	16-pin DIP array of 7 high-voltage, high-current NPN Darlington pairs with common suppression diodes	Driving unmounted stepper motors, relays, and solenoid coils	**5 Units**	**Consumable**
21	**Assorted Resistor Kit (1/4W Carbon Film)**	30 resistors each across 20 standard values: 10 Ω, 100 Ω, 220 Ω, 330 Ω, 470 Ω, 1 kΩ, 4.7 kΩ, 10 kΩ, 100 kΩ, 1 MΩ, etc.	Current-limiting for LEDs, pull-up/pull-down, and voltage dividers	**5 Kits (3,000 pcs)**	**Consumable**
22	**Assorted Capacitor Kit (Ceramic & Electrolytic)**	Assorted values: 22 pF, 100 pF, 10 nF, 100 nF (104), 1 µF, 10 µF, 100 µF, 1000 µF; 16V–50V	Noise filtering, crystal loading, decoupling & timing circuits	**5 Kits (1,000 pcs)**	**Consumable**
23	**Diodes and Transistors Kit**	BC547 / 2N3904 (NPN), BC557 / 2N3906 (PNP), 1N4148 (Fast signal), 1N4001 / 1N4007 (1A Rectifier)	Amplification, logic inverter switching, rectification & flyback protection	**3 Kits (75 pcs)**	**Consumable**
24	**Zener Diodes Assortment (3.3V, 5.1V, 12V)**	0.5W axial leaded Zener voltage reference diodes	Voltage clamping, regulation & basic 5V-to-3.3V logic level shifting	**3 Packs (30 pcs)**	**Consumable**
25	**Through-Hole Diffused LEDs (5mm) — Red**	Standard 5mm round diffused red LED; forward voltage 1.8V - 2.2V; current 20 mA	Visual output indicators and logic state signals	**5 Packs (250 pcs)**	**Consumable**
26	**Through-Hole Diffused LEDs (5mm) — Green**	Standard 5mm round diffused green LED; forward voltage 2.0V - 2.4V; current 20 mA	Power/Go status indicators	**5 Packs (250 pcs)**	**Consumable**
27	**Through-Hole Diffused LEDs (5mm) — Blue**	Standard 5mm round diffused blue LED; forward voltage 2.8V - 3.2V; current 20 mA	Special indicator and primary RGB color mixing	**5 Packs (250 pcs)**	**Consumable**
28	**Through-Hole Diffused LEDs (5mm) — Yellow**	Standard 5mm round diffused yellow LED; forward voltage 2.0V - 2.4V; current 20 mA	Warning and caution state signaling	**5 Packs (250 pcs)**	**Consumable**
29	**Through-Hole Clear LEDs (5mm) — White**	Ultra-bright 5mm clear white LED; forward voltage 3.0V - 3.4V; current 20 mA	Illumination and miniature flashlight projects	**5 Packs (250 pcs)**	**Consumable**
30	**RGB LEDs (5mm Common Cathode, 4-Pin)**	4-pin through-hole 5mm diffused RGB LED; common ground pin *(Requires 3 individual resistors)*	Color blending, PWM color mixing & status multi-color alerts	**3 Packs (60 pcs)**	**Consumable**

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31	**Tactile Momentary Push Buttons (6×6mm)**	4-pin breadboard-compatible momentary micro push buttons with colored caps	Manual digital inputs, reset triggers & interface buttons	**3 Packs (150 pcs)**	**Consumable**
32	**Miniature SPDT Slide Switches**	3-pin through-hole sub-miniature slide switches; ON-ON 2-position rating 0.5A 50V DC	Power master switching and hardware mode selection	**2 Packs (40 pcs)**	**Consumable**
33	**DIP Switch Array (4-Position & 8-Position)**	Standard 2.54mm pitch breadboard rocker/slide DIP switch blocks	Binary address setting, mode configuration & logic truth table tests	**5 Packs (20 pcs)**	**Consumable**
34	**Bare Tilt Ball Switch Sensor (SW-520D)**	2-pin brass-cased rolling ball tilt sensor switch	Simple gravity/tilt orientation detection without software libraries	**5 Units**	**Consumable**
35	**Bare Electret Microphone Capsule (9×7mm)**	2-pin raw electret microphone cartridge with solder pads *(Requires pull-up & coupling capacitor)*	Building custom acoustic sensor & sound trigger circuits	**5 Units**	**Consumable**
36	**Piezoelectric Transducer Disc Plate**	40mm copper-backed piezoelectric ceramic disc element with pre-soldered positive/negative wire leads	Vibration energy harvesting, knock detection & contact microphones	**5 Units**	**Consumable**
37	**Precision Analog Temperature Sensor (LM35)**	Precision linear analog temperature sensor; TO-92 3-pin transistor package; 10 mV/°C	Raw analog voltage temperature reading	**5 Units**	**Consumable**
38	**Force Sensitive Resistor (FSR)**	1.75× 1.5 inch square polymer thick-film pressure sensor; resistance varies with mechanical force	Pressure sensing, electronic scales & footstep pads	**3 Units**	**Consumable**
39	**Flexion Bend Sensor**	2.2-inch flexible carbon-resistive bend sensor; nominal flat resistance 25 kΩ	Robotic finger joint tracking and wearable glove controls	**3 Units**	**Consumable**
40	**Miniature Active Buzzer — Small**	5V DC active piezoelectric buzzer with internal oscillator (2-pin through-hole)	Audible alarms and beep notifications	**5 Units**	**Consumable**
41	**Loud Active Buzzer — Big**	5V–12V DC high-decibel active buzzer (2-lead round casing, 85+ dB output)	Perimeter intrusion alarms and fire warning buzzers	**5 Units**	**Consumable**
42	**Miniature Haptic Vibrating Motor**	10g coin/cylindrical vibration motor (1.5V - 3.7V DC) *(Requires transistor & flyback diode)*	Silent tactile alerts and wearable haptic feedback	**5 Units**	**Consumable**
43	**IR Optical Sensor Pair (TSOP1738 + 940nm LED)**	38 kHz infrared receiver IC (3-pin) paired with 940nm high-output IR emitting LED (2-pin)	Remote control decoders, IR beam breaks & optical tachometers	**5 Pairs**	**Consumable**